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4.0 Linux Text Editors

Text editors are useful utilities in any operating system as this can be used to make documents or edit configuration files. The Linux system provides two text editors: vi and nano. The vi text editor is a standard editor not just in Linux but also in any Unix version. Novice users may find the vi text editor hard to use at first since there are no "menus" on the screen. Users of vi text editors should know the vi modes, commands and short cuts to use it properly. Although vi maybe hard to use, most Unix/Linux users learn it because it is a more universal text editor than nano. The nano text editor is not necessarily available on Unix systems. To be clear, vi is actually vim in the Linux system. The vim text editor is the improved version of vi.

The nano text editor is simple and easy to use text editor in the command line interface of the Linux system. Unlike vi text editor, short cuts for text editor are found at the bottom of the screen. Most novice users prefers nano over vi text editor. Most short cuts are used with the meta key plus a letter key. An example short cut for saving is pressing "`ctrl + o`". The objectives of this section are:

- Be able to use the vi text editor to create or modify text files
- Be able to use the nano text editor to create or modify text files

4.1. The vi Text Editor

The vi text editor is a built-in in the Unix operating system. This text editor is a standard on all Unix versions. Only the most basic command set will be tackled in this section. A cheat sheet for vi text editor can be found at: http://www.atmos.albany.edu/daes/atmclasses/atm350/vi_cheat_sheet.pdf

vi has three modes of operation: command, entry mode and last-line mode. In the command mode, vi accepts command for cursor positioning, going into other modes, deleting characters or lines and searching for string patterns. The entry mode allows editing of text while the last-line mode allows saving and exiting from vi. Upon starting of vi, the program goes into the command mode. A summary of the commands is shown in Table 4.1.

Table 4.1 Summary of vi text editor commands

Command	Description
`<esc>`	Goes into command mode
`:`	Goes into last-line mode
`i or a`	Goes into entry mode. The "i" allows insertion of text before the cursor while "a" allows insertion of text after the cursor.
`h, j, k, l`	Moves cursor left, down, up, and right respectively. Can only be used in the command mode. Alternatively, cursor keys can be used as well but not standard on all Unix versions.
`O`	Adds a new line below the cursor. Can only be used in the command mode.
`x, dw, dd`	Deletes a character, a word, and a line where the cursor is, respectively. Also, adding a numeric before commands `dw` or `dd` will delete the corresponding number of words or lines, respectively. Can only be used in the command mode.
`yy, p, P`	The command yanks a line and put into clipboard, puts the contents of the clipboard below the cursor and puts the contents of the clipboard above the cursor, respectively. Can only be done in the command mode.
`/string, ?string`	Searches a string pattern forward in file and backward in a file. Can be only used in the command mode.
`:w, :w <filename>`	Saves the file
`:wq`	Exits vi and saves the file. Only in command mode.

`:q!`	Exits vi without saving the file. Only in command mode

NOTE: Do NOT use the numeric keypad on your keyboard if you have it. It has a different behavior from using the number keys between the function and letter keys.

Step 1 At the command prompt, enter the command "`vi rfc2186`".

1a. In which mode is your text editor in right now? *(See section 4.1 if you need a guide)*

> The text editor is in command mode right now

1b. In the command mode of vi, type "`:23`". Write down the first three (3) words on the line you are currently at.

> The first 3 words are, "This document describes". By using the ":" command, it brings us to the last-line mode.

1c. Place your cursor on the first letter of the word "*describes*". Use the arrow keys to do this. If the arrow keys are not supported, use (h, j, k, l) keys. Type the "`a`" command. In which operation mode are you in now?

> The "a" command brings us into entry mode.

1d. Type the word "*cat*" and press the escape key. Where did the cursor allow you to enter text? Was it before or after the position of the cursor? In which mode are you in now after pressing the escape key?

> The text was entered after the cursor, Pressing the esc key brings us back to command mode.

1e. Type the "`i`" command. In which mode are you in now?

> The "i" command brings us to the entry mode.

1f. Type the word "*hello*" and press the escape key. Where did the cursor allow you to enter text? Was it before or after the position of the cursor?

> The text was entered before the cursor. The "a" and "i" command both go to entry mode, but "i" enters text before the cursor while, "a" enters text after the cursor

1g. Press the escape the key. In which mode are you in now?

> I am back in command mode.

1h. Type the command "`x`". What happened to the line of text you are at? What does the command do?

> It deleted a character of the line we are at. It deletes a character.

1i. Type the command "`2dd`". What happened to the line of text you are at? What does the command do?

> The command deleted 2 lines of text. This command deletes lines, the number before the command indicates how many lines to delete.

1j. Type the command "`yy`". Where there any changes? What did the command do?

> There were no visible changes. This command yanks a line and puts it into the clipboard.

1k. Type the "`p`" command. What happened to the line of text you are at? What did the command do?

> The line I copied earlier using "yy" is now pasted. This command pastes the line in our clipboard.

1l. Type the command "`:wq`". What happened to the text editor? What did the command do?

> It exits and saves the file. The text editor is closed and we are back in the command prompt

1m. Enter the command "`vi rfc2186`". Go to line 23. Did `vi` save your edited file?

> Yes vi was able to save the file. I can see the changes I made.

1n. Duplicate the line again using the "`yy`" and "`p`" commands. (See steps 1j-1k)

1o. Type the command "`:q!`". What happened to the text editor? What did the command do?

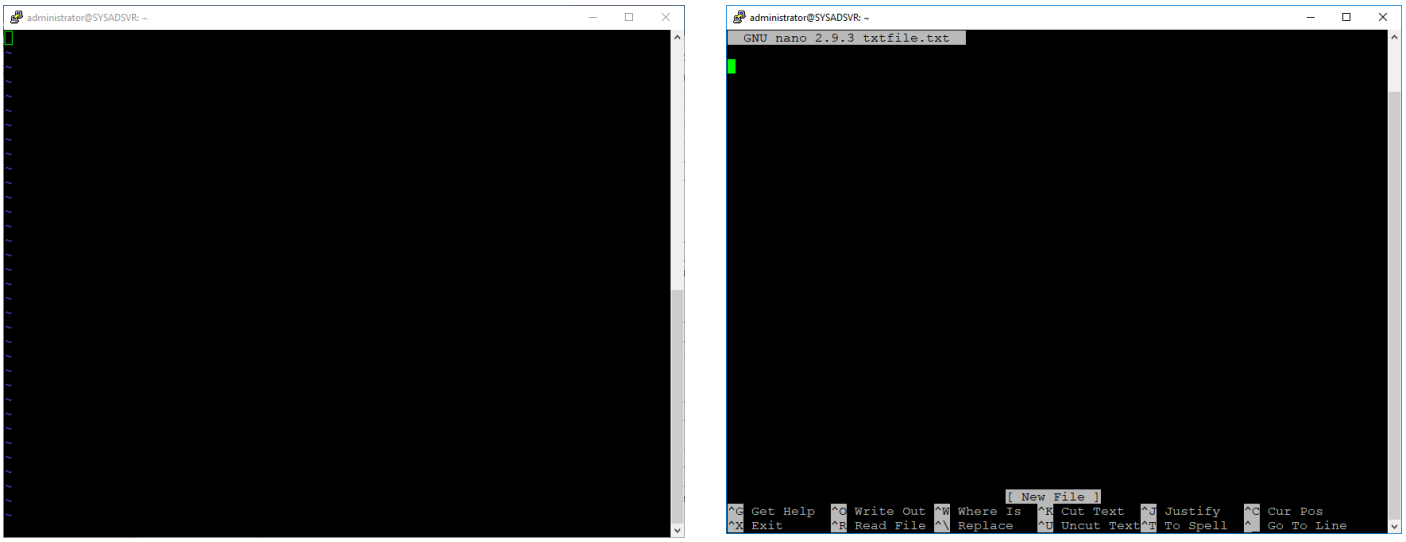
> The result is similar to ":wq", it exits the text editor. However, it does not save the changes we made

1p. Enter the command "`vi rfc2186`". Go to line 23. Did `vi` save the changes you made in step 1n?

> No, the text editor did not save the changes made.

4.2. The nano Text Editor

An alternative text editor on Linux is the nano. The nano text editor is simple and user friendly. Both text editors are shown in Fig. 4.1. It can be seen that the vi text editor does not show any short cut menu while the nano text editors show short cut menus at the bottom of the screen.



a. vi text editor on Linux b. nano text editor on Linux

Fig. 4.1 A side by side comparison of the vi (a) and nano (b) text editor

NOTE: Do NOT use the numeric keypad on your keyboard if you have it. It has a different behavior from using the number keys between the function and letter keys.

Step 2 At the command prompt, enter the command "`cp /home/files/rfc2186.txt rfc2186.txt`" and then type in "`nano rfc2186.txt`".

2a. Do you see the shortcut menu at the bottom of the screen?

Yes, there are shortcuts at the bottom of my screen

2b. In the command mode of nano, press "`CTRL`" + "_" then enter "23" when asked for the line number. Write down the first three (3) words on the line you are currently at.

The first 3 words are, "This document describes".

2c. Place your cursor on the first letter of the word "*describes*". Use the cursor keys to do this. Type in the word "*cat*". Were you able to edit the line?

Yes I am able to add the word 'cat' directly.

2d. Press the "`CTRL`" + "`x`" keys and then press "`y`" to confirm. Do not change the file name (just press enter when asked). Load the file again by typing in "`nano rfc2186.txt`". Was the document saved?

Yes the document was saved.

2e. Go to the line that was edited and insert the word "*hello*" after the word "*cat*". Press the "`CTRL`" + "`o`" keys and then press "`y`" to confirm. If not prompted, simply use the original filename. Did the nano text editor let you save the document? Did the nano text editor terminate after saving? What is the difference between the "`CTRL`" + "`o`" and "`CTRL`" + "`x`" shortcuts?

Yes the text editor saved the document. The main difference is that ctrl o saves the file without exiting out of the editor. While ctrl x saves the file and will exit out of the editor.

2f. Go to line 25 and press the "`CTRL`" + "`k`" keys. What happened to the line of text? What does the shortcut key do?

This key deletes a whole line. This shortcut deletes the line selected

2g. Go to next line and the press the "`CTRL`" + "`u`" keys. What happened to the line of text? What does the shortcut key do?

The line of text we just deleted is restored. I noticed that though it restores the deleted text, it is different from the usual ctrl-z because when I continued clicking ctrl-u the deleted text continued to be pasted again.

2h. Exit the nano text editor by pressing the "`CTRL`" + "`x`" keys. Press "`n`" when prompted if changes will be saved. What happened to the changes you made in 2f and 2g? (You can reopen the file to verify this)

The changes were not saved, this is because we chose not to save changes by typing 'n'.

4.3. Conclusion/Reflection

Between `vi` and `nano`, which text editor do you prefer using? List down at least one (1) thing you like and one (1) thing you dislike about your preferred text editor.

I prefer using nano. I like that nano has a shortcut menu because it is beneficial when you are new to text editors. It is convenient and easy to recall commands. I don't find this as a major inconvenience but I think it would be nice to see the commands I type when using the text editor. I believe this can avoid situations where you forget the commands you did and are confused of the current edits made.